

Solution Requirements

Date	26 June 2026
Team ID	XXXXXX
Project Name	RISING WATERS
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in using email and password.	High
2	Authorization levels	Admin manages flood data; users can only view alerts and maps.	High
		<i>services the system must connect to]</i>	

4	Transactions processing	System processes flood reports, water level updates, and sends alerts in real time	High
5	Reporting	Generates flood reports, affected area reports, and water level history.	Medium
6	Business rules, etc.	Alerts are sent only when water level crosses the danger threshold.	High
7	Compliance to laws or regulations	Follows data privacy and government disaster management guidelines.	Medium
8	<i>Other</i>	Supports emergency contacts	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System should quickly display flood data and alerts.	<i>Response time < 2 seconds</i>
2	Scalability	Supports increasing users during emergencies.	Up to 10000 concurrent users
3	Security & Data Privacy	Encrypts user data and secures login information.	<i>AES-256 encryption</i>
4	Reliability & Availability	System should be available during disasters.	99.9% uptime
5	Usability & Accessibility	Easy-to-use interface for all users.	Mobile-friendly and accessible
6	<i>Other</i>	Easy maintenance and future feature updates.	Modular and maintainable architecture